CS 3600 Project 1 Wrapper

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Due February 6th 2024 at 11:59pm EST via Canvas

Introduction

This Project Wrapper consists of a context paragraph, which identifies the topic of the wrapper, followed by four short-answer questions, each worth 1 point. Please limit your response to each question to a maximum of 200 words. Please write complete English sentences, but our focus is on the content of what you are writing and not your grammar. The goal of this assignment is to train your ability to reason through the consequences and ethical implications of computational intelligence. You should not focus on getting ”the right answer,” because the questions may be inherently open-ended or subject to interpretation. Instead, focus on demonstrating that you are able to consider the impacts of your AI design choices. **NOTE:** For this first wrapper, we have provided an answer for Question 1, to illustrate the length and quality of responses that we are looking for. This sample answer is around 100 words in length and will receive full credit (a free point for you!) Note that you are not required to answer this question.

Context

Consider a map of all of the roads in a city. A driver in this city is using a GPS app which locates the user's position on the map, and uses a A* implementation to identify an optimum route to the driver's destination using an admissible and consistent heuristic. Considering the intersections between roads to be the states, and the roads connecting the states to be the edges, denoting the possible actions, please answer the following questions.

Question 1

In the ice cream example in class, we used the length of roads as the edge cost between vertices (ice cream shops), and the resulting optimal route gave the shortest distance a car would have to travel by following the roads. How might we modify the search to account for speed limits? How might we account for traffic conditions if we know that traffic is flowing slower than the speed limit?

Example Answer (1 Free Point):

Speed limits and traffic affect time instead of distance, and so we can modify the edge costs to encode time. If there is an average speed va in miles/hour, then for an original graph edge cost d in miles, we can replace it with a new edge cost t = d/va in hours. We can now account for speed limits and traffic conditions by simply modifying the average speed for that edge to reflect the speed limit or a reduced average speed due to traffic conditions. As long as we have a consistent heuristic function, A* with the new costs will find the shortest duration path, which might not be the shortest distance path under the original cost model.

Question 2

Suppose there is a residential neighborhood where a lot of children live and play in the streets, which happens to be located between two very popular destinations. As more people use GPS-based route planning services, the neighborhood has started to see an increase in dangerously fast traffic. Suppose we wanted to discourage A* from routing cars through the neighborhood. What would happen if we artificially adjusted the speed limit on roads in the neighborhood versus if we artificially increased the heuristic values of intersections in the neighborhood? Would either approach guarantee that cars never cut through the neighborhood? Would either approach prevent people who live in the neighborhood from generating routes to and from their homes?

Answer:

The A* Search follows the heuristic where the next node is picked based off an f-value where f is comprised of a g-value, the cost to move from one node to the next node, and a h-value, the cost to move from a particular node to the final destination. Therefore, by adjusting the speed limits – or edge weights – the route through the neighborhood is less likely to be selected since the f-value is higher. This does not guarantee cars will never travel the neighborhood route, however, since congestion would create an artificial new speed limit on roads, thus potentially establishing the neighborhood road as the best route. Adjusting the heuristic values of intersections may cause longer routes to be prioritized even if they have a higher g-value. These longer routes – assumed to be interstates or less populated roads – may still cut through the neighborhood in certain circumstances, especially depending on the layout of the neighborhood. The question of if cars would cut through the neighborhood is dependent on the g-value and h-value; similarly, the values of g and h would ultimately dictate if people in the neighborhood can access their homes.

Question 3

There is currently a big societal concern regarding artificial intelligence and automation affecting jobs. How do route planning systems (such as Google Maps or Uber navigation) impact jobs? Is their impact mainly positive or mainly negative?

Answer:

The question of how jobs are impacted can be viewed from three primary angles. (1) Those who previously created maps or served in professions where it was unique to understand maps are out of business. (2) Mapping services may primarily route traffic along X-route, leaving businesses at Y-route with less traffic and thus less business. (3) Mapping services choose the best route using real-time heuristics and therefore may result in the employee being on time for work. The first and third perspectives are relatively niche in nature. In the first lens, these professions must accommodate with the changing times. In the third lens, I think a majority can view this as a positive, so there's not much room for unique commentary. The second perspective – on the surface – seems like a legitimate concern, however, loses its validity when it is realized that popular navigation services take into consideration how many users are rerouted as to preemptively avoid traffic. Furthermore, this concern is likely not a significant one since people's houses are likely spread out significantly enough such that the effect is mitigated. Ultimately, the effect of mapping services is positive as it considers from multiple heuristics and is not significantly biased.

Question 4

Reliance on artificial intelligence systems can change human behavior in unanticipated ways. Describe one way in which a route planning system can have an undesirable impact on human behavior.

Answer:

I feel I am obligated to state this answer is in no manner a reflection of myself. Nevertheless, I believe one of the primary instances in which AI coupled with route planning systems having an undesirable impact on human behavior is the estimated arrival time. Users – definitely not myself – may use the estimated arrival time as a challenge by which to beat or see to what extent they can beat it. Recently, this challenge has gotten more difficult with Waze's addition of a new AI feature that analyzes a user's personal driving style to provide with more accurate estimated arrival times. Users – again, not myself – may see this as an opportunity to still beat the estimated arrival time only this time, it is on *hard mode*.